

PUBLIC PORTFOLIO
PROJECT

JANUS QUANT RESEARCH

Point-in-time, leakage-aware quantitative research and paper-trading framework

Point-in-time

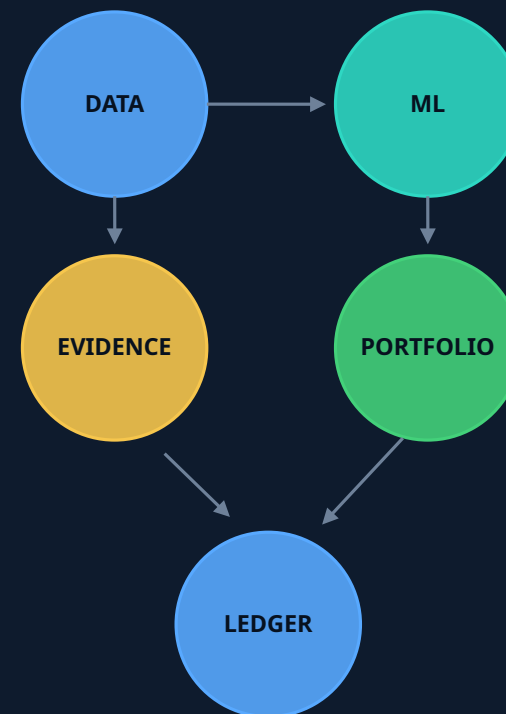
Conformal ML

HRP

Paper Ledger

Public portfolio snapshot · pre-S4 · September 2026

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github.com/aynacimehmet/janus-quant-research

A quant system should be correct before it is clever

Janus is built around four constraints that routinely break otherwise convincing backtests.

Point-in-time data

Every decision is bounded by what was actually observable at that timestamp.

Execution realism

Tax, FIFO lots, settlement, tradability and stale data are modeled in the ledger.

Probabilistic evidence

Quantile forecasts become calibrated evidence before they can influence allocation.

Separated gates

Technical readiness, strategy acceptance, paper operations and real capital are different decisions.

The project optimizes for trustworthy decision-making - not for the prettiest historical equity curve.

End-to-end research and paper-trading pipeline

One chain from observable data to a human-executable allocation proposal.



Execution layer

The same ledger semantics anchor backtests and paper trading: fills, tax, receivables, settlement, stale data and tradability are not post-hoc adjustments.

DuckDB + Parquet

Typer CLI

Shared Ledger

MLflow

pytest / ruff

launchd

The information boundary is explicit

Decision time, price formation, fill time and first investable return are intentionally separated.



Leakage rule

Future perturbations must not change past observations, actions, rewards or ledger state. If they do, the experiment is invalid.

Forecast → calibration → evidence → allocation

Each layer has a narrow role; no single score is allowed to jump directly into portfolio weights.

Quantile forecasting

Global LightGBM estimates q10 / q50 / q90 under rolling-origin walk-forward refits.

Hyperparameter selection

CPCV is used for forecast-model selection - not as a strategy backtest substitute.

Calibration

CQR plus an adaptive daily alpha path evaluates interval reliability over time.

Evidence selection

Conformal p-values are filtered with Benjamini-Hochberg FDR ($q = 0.20$).

Portfolio construction

HRP with Ledoit-Wolf covariance and concentration caps converts evidence into weights.

The cash proxy remains the default when the evidence set is empty.

The ledger is part of the model

Portfolio performance is defined by accounting state - not just by multiplying weights by returns.

State identity

$$V = \text{cash} + \text{receivables} + \text{units} \times \text{NAV}$$

- FIFO lots carry purchase date, cost basis and availability.
- Sales can create receivables before cash becomes spendable.
- Tax is realized on sale; stale data and suspension are distinct states.

Trade lifecycle



This lets Janus test portfolio rules under the same frictions that later appear in paper trading instead of applying costs after the fact.

Success is a sequence of independent gates

Forecast quality, calibration, strategy evidence and operational readiness are evaluated separately.

1

Forecast

Is the model better than naive baselines?

Rolling-origin walk-forward + CPCV for model selection; pinball loss and rank IC.

Forecast evidence

2

Calibration

Are uncertainty intervals trustworthy?

Time-block coverage, selected-set coverage, interval width and cash-rate diagnostics.

Coverage vs target

3

Strategy

Does the rule beat cash after tax?

Anchored selection period vs external test; 12-start sensitivity.

Post-tax excess vs B0

4

Selection risk

Did tuning just get lucky?

CSCV/PBO for strategy scans; Deflated Sharpe reported with trial count.

PBO + DSR

5

Operations

Can the process run reliably?

Paper trading, reconciliation, execution checks and separate real-capital approval.

Ops $\neq$ alpha

Research discipline is encoded in the software

The public snapshot is designed to be inspectable, reproducible and fail-closed.

Data engineering

DuckDB, immutable Parquet snapshots, source timestamps and point-in-time metadata.

ML / quant

LightGBM quantiles, conformal inference, Ledoit-Wolf HRP and regime research.

Execution engine

Shared ledger semantics for backtest and paper trading; FIFO, tax and settlement.

CLI & scheduling

Typer commands, incremental nightly flow and launchd automation on Apple Silicon.

Experiment tracking

Run identifiers, config hashes, data as-of metadata and MLflow integration.

QA discipline

400+ tests in the public snapshot, ruff, pre-commit, secret scanning and synthetic fixtures.

Fail-closed principle

Missing provenance, invalid temporal state or inconsistent execution metadata should stop the path instead of silently inventing a fallback.

AI agents accelerate the work; humans own the decisions

Janus uses a scoped, evidence-driven agent workflow for research engineering - not autonomous capital deployment.

Strategy & governance

ChatGPT / Gemini / Claude · strategy & review

- Task specifications, ADR drafts and acceptance criteria are developed across multiple AI workspaces.
- Temporal leakage, execution and validation assumptions are challenged adversarially.
- Final PASS / HOLD remains human-owned and evidence-based.

Scoped implementation

OpenCode · specialist agents + multi-model routing

- janus-quant / janus-data / janus-ml / janus-qa keep implementation scopes explicit.
- Model routing has included DeepSeek, Kimi K3 and GLM-5.3-Flash; GPT-6 Sol is used selectively as a stronger advisor when the task justifies it.
- Provider/model choice is operational, swappable and cost-aware rather than architectural.

Evidence & control

Human-in-the-loop release gates

- Repository contracts and accepted ADRs outrank model output.
- pytest, ruff, pre-commit and secret scans are release evidence.
- Real-data smoke is required where synthetic tests are not enough.

Operating principle

Least-privilege context, demand-driven reading, swappable/cost-aware model routing, and no silent contract changes. AI assists research engineering; it does not approve live capital.

ChatGPT

Gemini

Claude

OpenCode

DeepSeek

Kimi K3

GLM-5.3

GPT-6 Sol

GitHub

Several “reasonable” ideas were rejected after they failed harder tests

The project records not only what worked, but why plausible approaches were abandoned.

Lower-bound ranking looked statistically safe

WHAT FAILED

Conditioning on a favorable lower bound distorted coverage inside the selected set - a winner's-curse problem.

DECISION / LEARNING

Rejected as production evidence. Selection moved to conformal p-values + BH FDR.

Aggregate calibration looked acceptable

WHAT FAILED

A maturation-time mapping bug could keep adaptive alpha effectively frozen while broad diagnostics still looked plausible.

DECISION / LEARNING

Matched matured outcomes to their original historical decision and added explicit regression checks.

A successful technical sprint can feel “done”

WHAT FAILED

Passing tests says nothing by itself about strategy alpha, paper operations or real-capital readiness.

DECISION / LEARNING

Technical, strategy, pilot and capital gates are intentionally independent.

What the pre-S4 public snapshot actually contains

The repository intentionally stops before reinforcement-learning exposure control.

IMPLEMENTED

- Point-in-time ingestion and feature pipeline
- Quantile LightGBM walk-forward forecasts
- CQR / adaptive-alpha calibration
- Conformal p-values + BH FDR selection
- HRP allocation and concentration controls
- Execution-aware backtest + paper ledger
- CLI, incremental pipeline, QA and documentation

FUTURE RESEARCH

- Adaptive exposure control / reinforcement learning (S4)
- US equity leg and market-specific execution model
- Controlled lineage / promotion for full rebuilds
- Real-capital approval only after separate operational and assumption gates

Public snapshot = research and paper-trading framework. It is not an investment product and does not constitute investment advice.

PORTFOLIO TAKEAWAY

A compact example of financial ML + systems thinking

Janus demonstrates how I approach quantitative research when the difficult part is not only prediction, but information timing, uncertainty, execution accounting, validation and operational safety.

Financial ML

Point-in-time data

Quant research

Execution modeling

Agentic AI

MLOps / QA



[github.com/aynacimehmet
/janus-quant-research](https://github.com/aynacimehmet/janus-quant-research)

Scan to review the public source, tests and
methodology.

Public portfolio project · educational / research use · no investment advice.